

Presentation of the week

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SNORD AND TRNA FAMILY

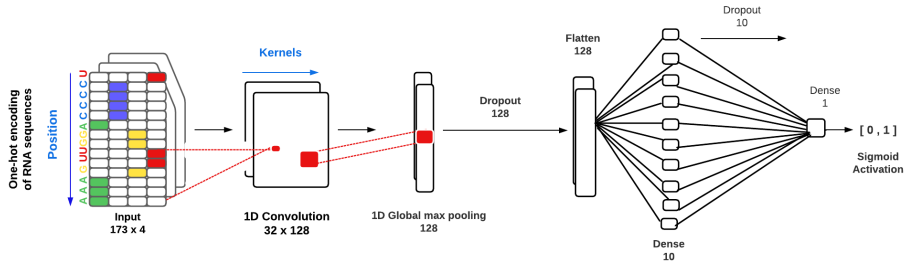


Figure – Model architecture for RNA Sequences

SNORD AND TRNA FAMILY

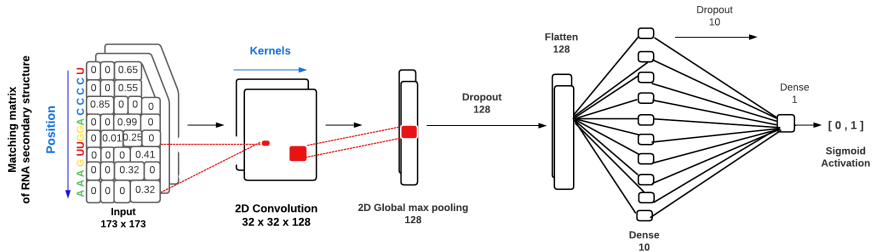


Figure – Model architecture for RNA Secondary structures

SNORD AND TRNA FAMILY

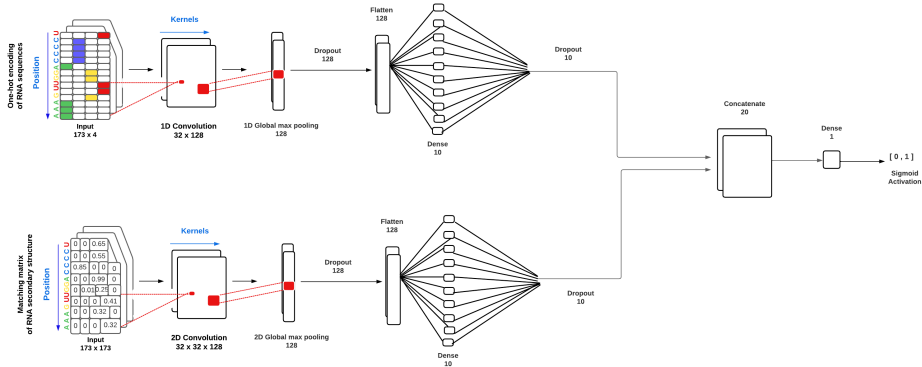


Figure – Model architecture with two inputs

SNORD AND TRNA FAMILY

Confusion Matrix

Model for RNA Sequences			
Actual		Predicted	
		SNORD	TRNA
	SNORD	863	0
	TRNA	3	888

Model for RNA Secondary Structures			
Actual		Predicted	
		SNORD	TRNA
	SNORD	836	27
	TRNA	72	819

Model with Two Inputs			
Actual		Predicted	
		SNORD	TRNA
	SNORD	863	0
	TRNA	3	88

SNORD AND TRNA FAMILY

Metrics	Model for RNA Sequences	Model for RNA Secondary structures	Model with two inputs
Error rate	0.0017	0.0564	0.0017
Accuracy	0.9983	0.9436	0.9982
Recall	0.9966	0.9192	0.9966
Precision	1	0.9681	1
F1 score	0.9983	0.9430	0.9983
Cohen's kappa	0.9966	0.8872	0.9966
ROC AUC	0.9983	0.9439	0.9983

Table – Performance metrics for different models

HACA AND U1 FAMILY

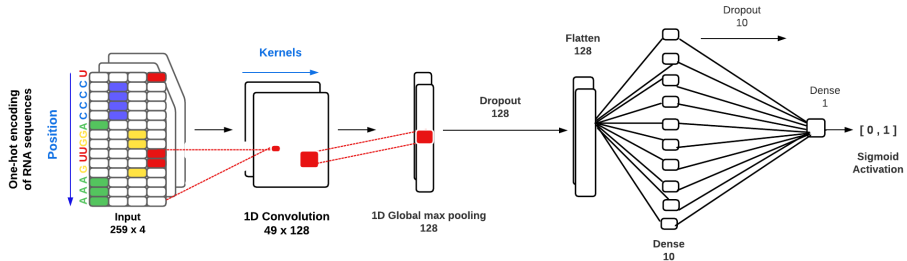


Figure – Model architecture for RNA Sequences

HACA AND U1 FAMILY

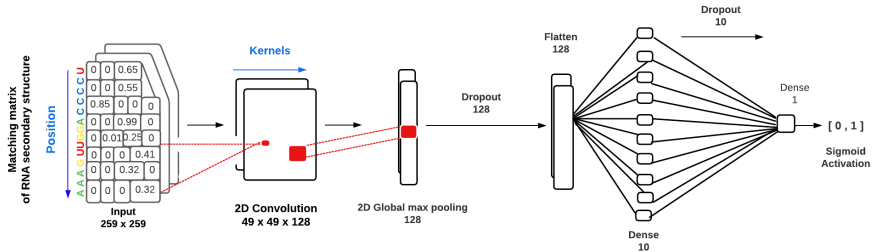


Figure – Model architecture for RNA Secondary structures

HACA AND U1 FAMILY

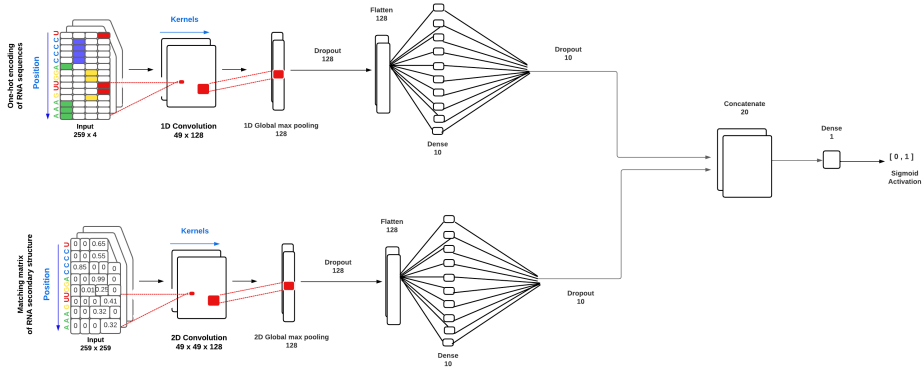


Figure – Model architecture with two inputs

Confusion Matrix

Model for RNA Sequences			
Actual		Predicted	
		HACA	U1
	HACA	1582	0
	U1	2	1590

Model for RNA Secondary Structures			
Actual		Predicted	
		HACA	U1
	HACA	1517	65
	U1	112	1480

Model with Two Inputs			
Actual		Predicted	
		HACA	U1
	HACA	1582	0
	U1	0	1592

HACA AND U1 FAMILY

Metrics	Model for RNA Sequences	Model for RNA Secondary structures	Model with two inputs
Error rate	0.0006	0.0558	0
Accuracy	0.9994	0.9442	1
Recall	0.9987	0.9296	1
Precision	1	0.9579	1
F1 score	0.9994	0.9436	1
Cohen's kappa	0.9987	0.8885	1
ROC AUC	0.9994	0.9443	1

Table – Performance metrics for different models

SNORD , TRNA , HACA AND U1 FAMILY

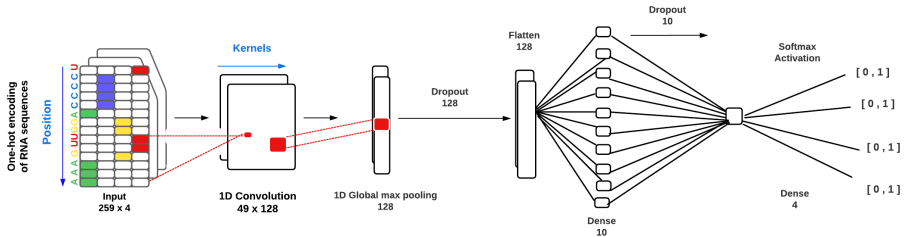


Figure – Model architecture for RNA Sequences

SNORD , TRNA , HACA AND U1 FAMILY

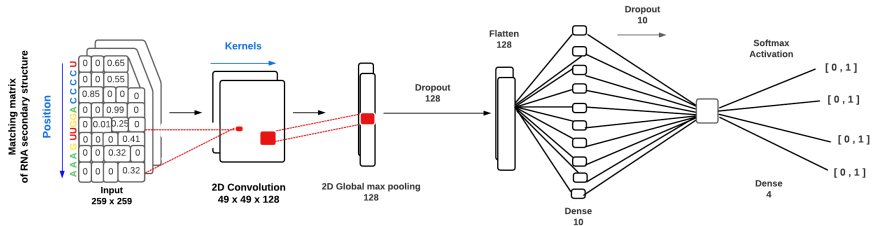


Figure – Model architecture for RNA Secondary structures model

SNORD , TRNA , HACA AND U1 FAMILY

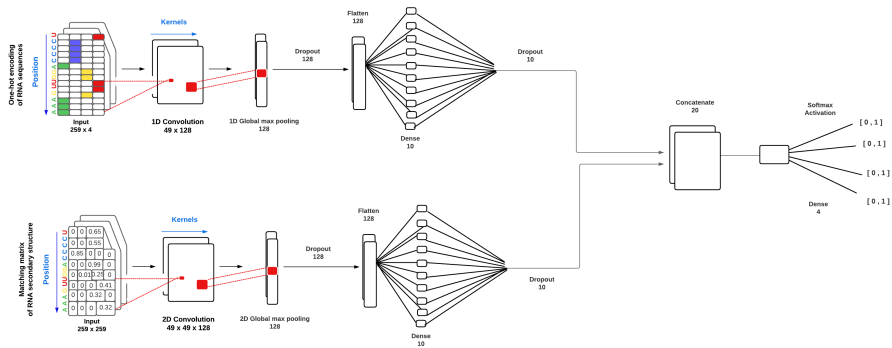


Figure – Model architecture with two inputs

SNORD , TRNA , HACA AND U1 FAMILY

	Precision	Recall	F1-Score	Support
SNORD	1.00	1.00	1.00	877
TRNA	1.00	0.99	1.00	877
HACA	0.99	1.00	1.00	1587
U1	1.00	1.00	1.00	1587
Accuracy : 1.00, Support : 4928				
Macro avg : 1.00, Support : 4928				
Weighted avg : 1.00, Support : 4928				
Cohen's Kappa : 0.995825				
ROC AUC : 0.998616				

	SNORD	TRNA	HACA	U1
SNORD	873	1	3	0
TRNA	1	872	3	1
HACA	1	0	1585	1
U1	1	1	2	1583

Table – Classification Report and Confusion Matrix for RNA Sequences model

SNORD , TRNA , HACA AND U1 FAMILY

	Precision	Recall	F1-Score	Support
SNORD	0.78	0.88	0.83	877
TRNA	0.95	0.88	0.91	877
HACA	0.89	0.88	0.88	1587
U1	0.96	0.94	0.95	1587
Accuracy : 0.90, Support : 4928				
Macro avg : 0.89, Support : 4928				
Weighted avg : 0.90, Support : 4928				
Cohen's Kappa : 0.862524				
ROC AUC : 0.930850				

	SNORD	TRNA	HACA	U1
SNORD	772	31	65	9
TRNA	78	773	23	3
HACA	129	13	1393	52
U1	6	0	86	1495

Table – Classification Report and Confusion Matrix for RNA Secondary structures model

SNORD , TRNA , HACA AND U1 FAMILY

	Precision	Recall	F1-Score	Support
SNORD	1.00	1.00	1.00	877
TRNA	1.00	1.00	1.00	877
HACA	1.00	1.00	1.00	1587
U1	1.00	1.00	1.00	1587
Accuracy : 1.00, Support : 4928				
Macro avg : 1.00, Support : 4928				
Weighted avg : 1.00, Support : 4928				
Cohen's Kappa : 0.998330				
ROC AUC : 0.999131				

	SNORD	TRNA	HACA	U1
SNORD	876	0	1	0
TRNA	1	875	1	0
HACA	1	1	1585	0
U1	0	0	1	1586

Table – Classification Report and Confusion Matrix for model with two inputs

SNORD AND TRNA FAMILY WITH DNABERT MODEL

Error rate	0.1267
Accuracy	0.8732
Recall	0.8187
Precision	0.9077
F1 score	0.8609
Cohen's kappa	0.7450
ROC AUC	0.9110

Table – Performance metrics for RNA sequences model